

Task 1 — BookStore SQL Database

User Stories

As the owner of an online bookstore, I want to:

- Sell books to my customers and keep records of every sale.
- Know which books are running low in stock.
- Organize my books in a way that helps customers find what they want.
- See which books sell the most and which customers buy the most.

Customers should be able to:

- Browse books by topic and by the person who wrote them.
- Buy more than one book in a single purchase.

Requirements

- All data must stay clean and consistent over time.
- Prices cannot be zero or negative.
- Stock cannot be negative.
- The same email cannot be registered for two different customers.
- Removing any item should not break old sales records.
- Old purchases must keep their original price even if the book price changes later.

Tasks

1. Design the database schema based on the stories and requirements above. Decide the tables, columns, keys, and relationships on your own.
2. Insert sample data that covers all the scenarios. Use enough data so your queries return meaningful results.
3. List all books sorted by price from highest to lowest.
4. Show book titles in uppercase and author names in lowercase.
5. Show every book with its category and its author.
6. Show every customer with the number of purchases they have made.
7. List the top 5 best-selling books.
8. Find the city with the highest number of customers.
9. List categories that have more than 5 books.
10. Find all books that cost more than the average book price.
11. Find customers who have never made a purchase.
12. Show the total revenue for each month.
13. Create a view that combines book title, category, author, and price.

14. Create a stored procedure that returns all purchases of a given customer with the totals.

Submission Rules

- One SQL file in a public GitHub repo, with answers numbered to match the tasks above.
- README must contain the ERD and a short description.